Machine Learning and Pattern Recognition

Laboratory 04: Probability and Density Estimation

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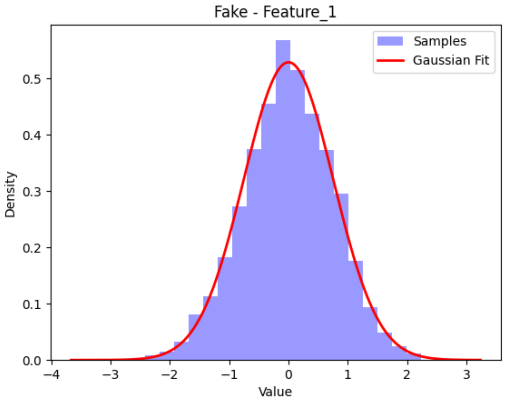
May 15, 2026

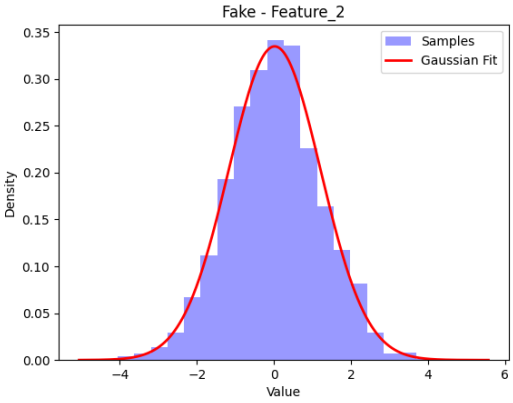
1D Gaussian Density Estimation

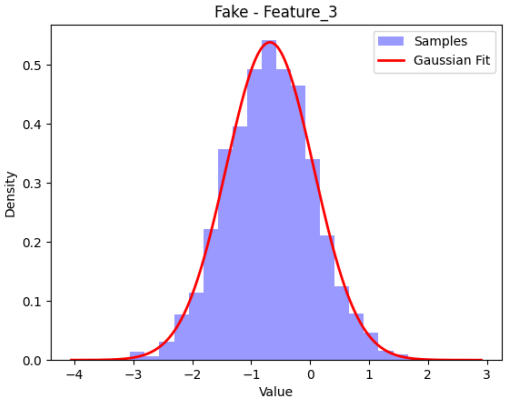
Features with a Good Gaussian Fit

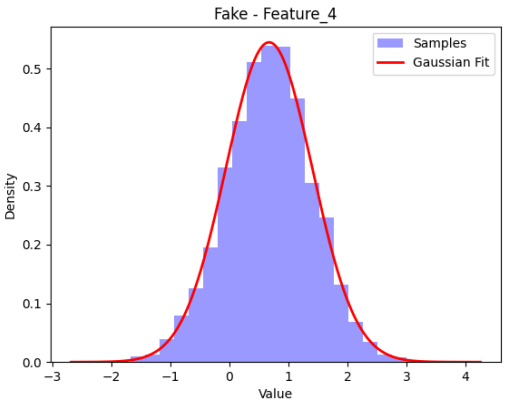
The Maximum Likelihood (ML) 1D Gaussian model provides a good, accurate fit for features that exhibit symmetric, bell-shaped distributions. For these features, the empirical data closely follows the theoretical exponential decay from the mean dictated by the computed variance.

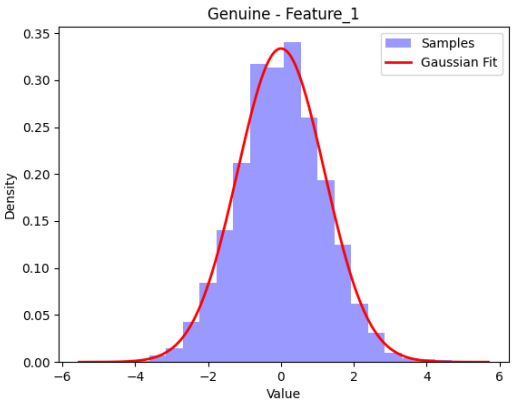
- **Fake Class:** Features 1, 2, 3, and 4. The histograms show a clear, symmetric distribution around their respective means (e.g., Feature 1 has a mean of 0.0029 and a variance of 0.5696, and visually matches the theoretical curve well).
- **Genuine Class:** Features 1, 2, 3, and 4. These features present a strong symmetric distribution around their empirical means. For instance, Feature 1 (mean 0.0005, variance 1.4302) and Feature 2 (mean -0.0085, variance 0.5783) are well-approximated by the Gaussian bell curve without severe skewness.

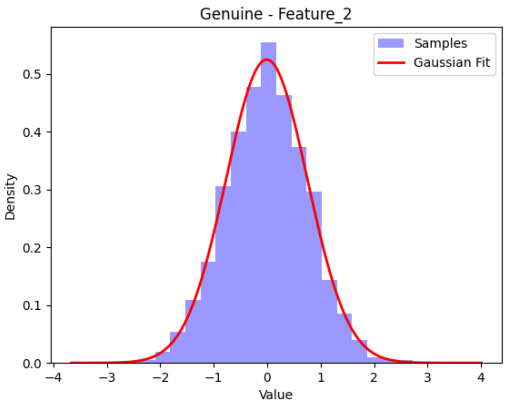

Fake - Feature_1
Samples
Gaussian Fit
Density
Value

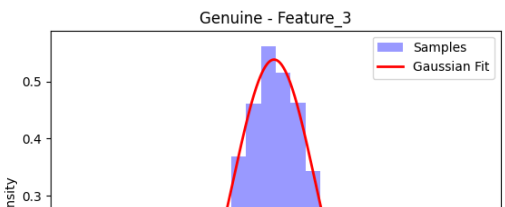

Fake - Feature_2
Samples
Gaussian Fit
Density
Value

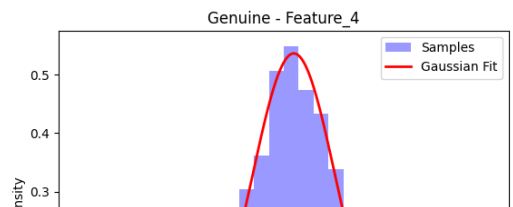

Fake - Feature_3
Samples
Gaussian Fit
Density
Value


Fake - Feature_4
Samples
Gaussian Fit
Density
Value


Genuine - Feature_1
Samples
Gaussian Fit
Density
Value


Genuine - Feature_2
Samples
Gaussian Fit
Density
Value


Genuine - Feature_3
Samples
Gaussian Fit


Genuine - Feature_4
Samples
Gaussian Fit

Features with an Inaccurate Gaussian Fit

The Gaussian model is highly inaccurate for features that exhibit strong asymmetry (skewness), multiple modes, or sharp drop-offs. The strict mathematical constraints of the Gaussian model (perfect symmetry around the mean) fail to capture these irregular empirical geometries.

- **Fake Class:** Features 5 and 6 are heavily skewed to the left. For instance, despite Feature 5 having a computed ML mean of 0.0280 and variance of 0.6801, the visual data has sharp right-sided peaks and long left tails that the symmetric curve overestimates or misses entirely.
- **Genuine Class:** Features 5 and 6 exhibit severe negative skewness. The rigid symmetry of the Gaussian model fails here because it predicts high probabilities in regions on the right side where empirical samples are virtually nonexistent, while underestimating the heavy tails on the left.

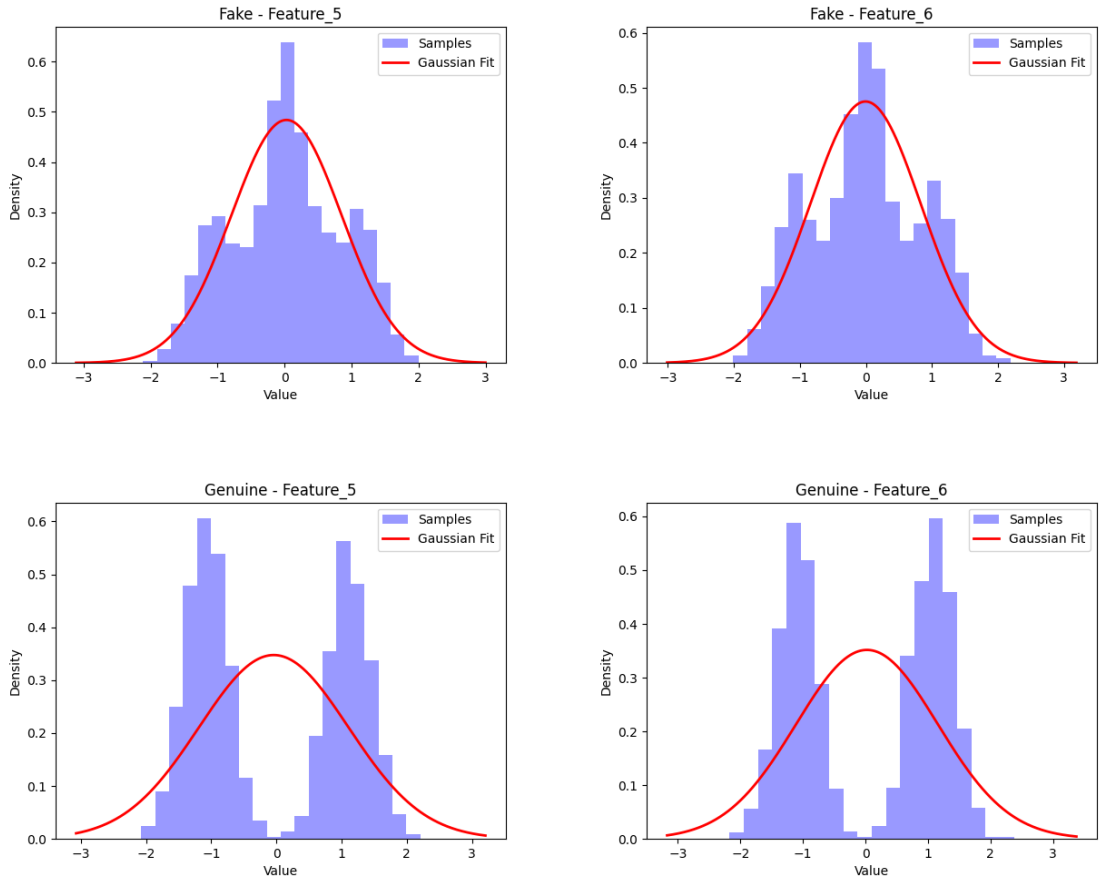


Figure 2: Features exhibiting a highly inaccurate fit due to skewness and sharp drop-offs.

This visual analysis indicates that while a Gaussian classifier is mathematically elegant and computationally fast, its rigid structural assumptions will likely cause classification errors when evaluating test samples based on these heavily skewed features.